A choice experiment for testing stochastic discrete choice conditions

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Abstract

We run an experiment covering 32 different choice domains, including fine art, cars and pizza. For each domain, there is a set of five related items, and 26 menus—the subsets of two or more items. We measure the probability of choosing each item from each menu, at the population level. We do this by presenting, within each domain, each of its 26 menus to a distinct group of at least 40 participants (1042 participants in total) and observing each participant's choice of one option. Each participant faces exactly one menu per domain. The purpose of the experiment is to allow tests of various conditions over choice probabilities, over a wide variety of choice domains. Such conditions include constraints implied by classes of choice models and axioms for probabilistic choice. The experiment is particularly well suited for testing conditions, like regularity, the random utility condition and the multiplicative inequality, which constrain choice probabilities across all menus of a domain. For each domain and each of the three conditions above, we report Bayes factors giving evidence for or against the condition. Regularity and random utility, conditions that most discrete choice models satisfy, survive this battery of tests: across the 32

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domains, Bayes factors generally support them and nowhere is there more than slight evidence against them. In contrast, very strong evidence against the multiplicative inequality in three domains shows that it cannot be considered as a universal condition on population-level choice. Classes of models implying the multiplicative inequality—including the classes of independent random utility models and Elimination by Aspects models—are not universally adequate for modelling population-level choice.

Key words: Discrete choice, Stochastic choice, Choice experiment, Context effects, Random utility, Bayesian inference

JEL codes: C11, C12, C15, C35, C91, D01, D90.

1 Introduction

Discrete choice models, which describe the choices of decision makers from finite menus of options, are used extensively in economics, psychology, marketing, transportation research, health research and other disciplines. Most empirical applications seek to measure how attributes of choice options affect choice probabilities in particular cases; for example, how commute times and other features affect the probabilities of choosing various transportation options (e.g. Train, 1978). Here we contribute to a distinct and more abstract research agenda that tests various conditions on choice probabilities, without regard to item characteristics, across a wide variety of choice environments. Specifically, we contribute new experimental data particularly suited to this endeavour.

Consider the following choice experiment, one of 32 we performed using an internet survey. We asked each participant "Which one of the following cities do you think is furthest north?" and then presented them with one *menu*, a list of at least two options taken from the following set of five cities: Warsaw, Poland; London, United Kingdom; Vancouver, Canada; Paris, France; and Seattle, United States. The experimental manipulation consisted of varying the menu across participants. There are 26 possible menus from the set of five cities and we presented each menu to a distinct group of 40 or 41 participants (1042 participants in total). Table 1 shows the choice counts we observed, by menu and item.

The following examples illustrate several themes of the paper. From the menu of Vancouver and Paris, 17 out of 40 participants chose Vancouver and 23 chose Paris; from the menu of Vancouver, Paris and Seattle, 26 out of 40 chose Vancouver; 14, Paris; and none, Seattle. It appears that adding Seattle to the menu of Vancouver and Paris increases the probability that participants choose Vancouver. This would be a violation of a condition on choice probabilities known as

	Choice counts					Choice proportions				
Menu A	$n_A(a)$	$n_A(b)$	$n_A(c)$	$n_A(d)$	$n_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	22	18	-	-	-	0.550	0.450	-	-	-
$\{a, c\}$	31	-	9	-	-	0.775	-	0.225	-	-
$\{b, c\}$	-	29	11	-	-	-	0.725	0.275	-	-
$\{a, b, c\}$	17	16	7	-	-	0.425	0.400	0.175	-	-
$\{a, d\}$	25	-	-	16	-	0.610	-	-	0.390	-
$\{b, d\}$	-	32	-	8	-	-	0.800	-	0.200	-
$\{a, b, d\}$	24	14	-	2	-	0.600	0.350	-	0.050	-
$\{c, d\}$	-	-	17	23	-	-	-	0.425	0.575	-
$\{a, c, d\}$	30	-	6	4	-	0.750	-	0.150	0.100	-
$\{b, c, d\}$	-	24	14	2	-	-	0.600	0.350	0.050	-
$\{a, b, c, d\}$	23	4	12	1	-	0.575	0.100	0.300	0.025	-
$\{a, e\}$	30	-	-	-	10	0.750	-	-	-	0.250
$\{b, e\}$	-	31	-	-	9	-	0.775	-	-	0.225
$\{a, b, e\}$	19	15	-	-	6	0.475	0.375	-	-	0.150
$\{c, e\}$	-	-	36	-	4	-	-	0.900	-	0.100
$\{a, c, e\}$	30	-	9	-	2	0.732	-	0.220	-	0.049
$\{b, c, e\}$	-	22	15	-	3	-	0.550	0.375	-	0.075
$\{a, b, c, e\}$	9	15	14	-	2	0.225	0.375	0.350	-	0.050
$\{d, e\}$	-	-	-	26	14	-	-	-	0.650	0.350
$\{a, d, e\}$	26	-	-	5	9	0.650	-	-	0.125	0.225
$\{b, d, e\}$	-	25	-	8	7	-	0.625	-	0.200	0.175
$\{a, b, d, e\}$	16	18	-	3	3	0.400	0.450	-	0.075	0.075
$\{c, d, e\}$	-	-	26	14	0	-	-	0.650	0.350	0.000
$\{a, c, d, e\}$	33	-	4	1	2	0.825	-	0.100	0.025	0.050
$\{b, c, d, e\}$	-	24	13	3	0	-	0.600	0.325	0.075	0.000
$\{a, b, c, d, e\}$	19	9	10	2	0	0.475	0.225	0.250	0.050	0.000

Table 1: Observed choice counts and proportions for the "Latitude" domain, where the prompt is "Which one of the following cities do you think is furthest north?" and the universe of items is $\{a, b, c, d, e\}$, where a is Warsaw, Poland; b is London, United Kingdom, c is Vancouver, Canada; d is Paris, France; and e is Seattle, United States. Each row gives counts and proportions for the menu shown in the first column. The second through sixth columns show the number of participants who chose the items a, b, c, d and e, respectively, from the menu. $n_A(x)$ is the number of participants who chose item x from menu A, $x = a, b, c, d, e$. Where an item is not in the menu, a dash is displayed. Thus, a zero unambiguously means that an item was available but never chosen from the menu in question. The final five columns show the corresponding choice proportions. $\hat{P}_A(x)$ is the proportion of participants who chose item x from menu A. Analogous tables for all 32 domains are provided in the supplementary materials.

regularity, which holds that if you add an option to a menu, the probability of choosing an existing option cannot increase.

Regularity is one of many stochastic choice conditions that have been studied in the discrete choice literature. It is easy to understand and has some intuitive appeal as a plausible axiom of choice. We will see later that it is a condition that is implied by a large class of choice models. Regularity can be regarded as a hypothesis and tested directly, allowing us to draw conclusions about this class of models without estimating model parameters. It consists of a set of inequalities that jointly constrain choice probabilities across all 26 menus of the five-item universe described above. Finally, regularity is an abstract condition: it makes no reference to attributes of various options and can therefore be applied across very different choice environments.

Regularity, and other conditions implied by various classes of models, are targets of a large literature on *context effects*. This literature gathers empirical evidence against these conditions, and by extension, the models that imply them. The attraction effect, for example, occurs when adding an option that is clearly inferior to one option in a menu increases the probability of choosing the superior option, a clear violation of regularity. We designed the Latitude domain to include configurations of cities conducive to an attraction effect, based on our conjecture that it would be obvious to most participants, drawn from a panel of Canadian residents, that Vancouver is north of Seattle and would be seen as a clearly superior option. Indeed, from the menu consisting of these two cities, 36 chose Vancouver and 4 chose Seattle. The choice counts reported above reveal that when Seattle is added to the menu of Vancouver and Paris, the proportion of participants choosing Vancouver increased. This provides preliminary evidence for the attraction effect, but to assess the strength of the evidence, we must take into account sampling variation.

The main contribution of this paper is a publicly available dataset designed for testing abstract stochastic choice conditions, such as regularity, across a wide range of choice domains. The experiment includes 32 diverse choice domains, spanning qualitatively different kinds of choice, including consumer choice (cars, beer), aesthetic preference (colours, art), factual judgment (latitude, population), and perceptual discrimination (triangle areas). This variety allows us and others to discover properties of random choice behaviour that hold widely. Some domains are specifically designed to be conducive to the attraction effect or other context effects. We do so to stress-test conditions that are inconsistent with the effects and, more generally, to promote choice patterns that many consider anomalous. Measuring choice probabilities on all menus allows us to test, jointly, the complete set of implications of a condition, such as regularity, that constrains choice probabilities

over all menus. In this way, we can expose every implication of the condition to empirical scrutiny. Any unfavourable evidence is evidence against decision theories that imply the condition.

Section 2 develops the theoretical framework for choice probabilities. It documents three conditions on choice probabilities in detail, including regularity, and points to others in the literature. It discusses the applicability of population-level data to inferences about individual-level behaviour. It describes three context effects and outlines the motivation for designing some of the domains to favour them. Section 3 describes the experimental design in detail. Section 4 reviews the Bayesian models and inferential methods used in the empirical illustrations. These methods are adapted from earlier work but are summarized here to make the paper reasonably self-contained and reproducible. Section 5 illustrates some analyses that are possible using our data, including tests of the three conditions described in Section 2 and an example of testing for context effects. Section 6 contains statements of transparency and openness and describes where to find data, code, and documents related to the experiment. Section 7 concludes.

2 Framework

Here we set up the probabilistic choice framework by defining a *random choice model*. We define three conditions on choice probabilities we investigate in a later empirical section: regularity, the Block-Marschak inequalities and the multiplicative inequality. We describe versions of three context effects that motivated parts of our experimental design: the attraction, similarity and compromise effects. Finally, we discuss individual- and population-level choice experiments and identify an aggregation property relating conditions on individual choice to conditions on population choice.

2.1 Random choice models

A random choice model specifies choice probabilities for a set $U = \{a_1, a_2, \ldots, a_n\}$ of related choice items, which we call a universe. For each of the 32 choice domains in our experiment, there is a universe containing five related and easily comparable choice items. We will write $U = \{a, b, c, d, e\}$ for a generic five-item universe. For the "Latitude" domain of the experiment, where participants see the prompt question "Which one of the following cities do you think is furthest north?", a is Warsaw, b is London, c is Vancouver, d is Paris and e is Seattle.

Each participant sees a single menu from each domain and must choose a single option from each menu. Different participants see different menus. Since any menu may be presented, a random

choice model needs to specify choice probabilities for each option of each menu. For a given universe U, let A be any non-empty subset of U and x be any option in A. Define $P_A(x)$ as the probability of choosing x from A, and P_A as the vector of all choice probabilities on menu A. One of the 26 menus is $\{a, b\}$; $P_{\{a,b\}}(a)$ and $P_{\{a,b\}}(b)$ are the probabilities of choosing Warsaw and London from the menu consisting of these two cities and $P_{\{a,b\}}$ is the vector of both. We learn something about these probabilities from the experiment: 22 participants chose Warsaw and 18 chose London, as we can see in the first row of Table 1, which tabulates choice counts for each of the 26 menus of the "Latitude" domain. We also define, for observed choice counts, $\hat{P}_A(x)$ as the proportion of times that an item x is chosen from a menu A, as long as x is in A. For example, we compute the observed choice proportions $\hat{P}_{\{a,b\}}(a) = 0.55$ and $\hat{P}_{\{a,b\}}(b) = 0.45$ for the "Latitude" domain. Although we defined menus earlier as choice sets with at least two items, the notation $P_A(x)$ allows for choice sets A with a single item. This is to ensure that we can concisely write down conditions like the Block-Marschak inequalities below; for each item x in U, the choice probability $P_{\{x\}}(x)$ is trivially equal to one.

A random choice model for the universe U is the complete specification of all choice probabilities $P_A(x)$ for all menus A of U and all items x in A, and is denoted P. In the case of repeated trials, we assume that observed choices are statistically independent, both within and across menus. We also assume that for each menu, observed choices are identically distributed. Thus a random choice model P for a universe gives a complete probabilistic description of an observed sequence of choices from menus of that universe. In a manner analogous to the definition of P, we define $\hat{P}$ as the complete specification of all the choice proportions $\hat{P}_A(x)$ observed in an experiment. By construction, $\hat{P}$ satisfies the same non-negativity and adding-up constraints that P does. The term random choice model is often used to describe logit and other, often parametric, sets of random choice models. We will refer to these as *classes* of random choice models.

One particularly important class of random choice models consists of those that describe the behaviour of random utility maximizers. Each time such a decision maker encounters a menu in a choice trial, they draw, from some probability distribution, utility values for all items in the universe and then choose the option in that menu with the highest realized utility. Utility is drawn independently of which menu is being offered. From trial to trial, utilities are independent and identically distributed. Within a trial, however, the utilities of the various items may be statistically dependent. We refer to the condition that given choice probabilities agree with the choice probabilities of some random utility maximizer as *random utility*. For a precise definition as

well as a detailed exposition of random utility and how it relates to random choice models, using the same notation as this paper, see McCausland et al. (2020). Most classes of random choice models used in applied work satisfy the random utility condition, including the Logit and Probit classes.

2.2 Global testable conditions for stochastic choice

Our term "condition" covers any axiom, constraint, hypothesis, property, proposition or rule about probabilistic choice behaviour that can be expressed as a restriction over the various choice probabilities of a random choice model.

We have already seen regularity, the condition that if you add options to a menu, the probability of choosing an existing option does not go up. Formally, a random choice model P for a universe U of choice items satisfies regularity if and only if for all $x \in A \subseteq B \subseteq U$,

$$P_A(x) \geq P_B(x). \tag{1}$$

Luce & Suppes (1965) describe regularity and show that it is necessary for the random utility condition.

Recall that regularity constrains choice probabilities across all menus. In fact, we can claim something stronger. Every P_A is *pivotal*, in the sense that for any random choice model satisfying regularity and any menu A, one can change the vector P_A so that the modified model violates regularity. The pivotal role of every P_A is a compelling reason to collect choice data from every menu of a universe, if one wants to test regularity for some domain.

The next condition, the Block-Marschak inequalities, is relevant because it is not only necessary but also sufficient for the random utility condition. A random choice model P on a universe U satisfies the Block-Marschak inequalities if and only if for all non-empty $A \subseteq U$ and all $x \in A$,

$$\sum_{B\colon\, A \subseteq B \subseteq U} (-1)^{|B \backslash A|} P_B(x) \geq 0, \tag{2}$$

where $|B \backslash A|$ is the number of items in B that are not also in A. ($\backslash$ is the set difference operator and $|\cdot|$ denotes cardinality.) This condition is far less interpretable than regularity, although see Barberá & Pattanaik (1986, p. 713) for an interpretation of the sum in Equation (2) for given A and x, in the case that the full set of conditions holds. Block & Marschak (1960) show that the

Block-Marschak inequalities are necessary for the random utility condition and Falmagne (1978) shows they are sufficient. Given the equivalence of the random utility condition and the Block-Marschak inequalities, we will henceforth refer to the condition using the more concise and familiar term, random utility. For the random utility condition, as for regularity, each P_A is pivotal: since regularity is necessary for random utility, the same change in P_A that breaks regularity also breaks the random utility condition.

The third and final testable condition we consider is the multiplicative inequality, introduced by Sattath & Tversky (1976), who claim "it has some appeal as a principle of individual choice behaviour". It states that the probability of choosing an option x from a union $A \cup B$ of two menus is at least as great as the probability of choosing x from A and then again from B in independent trials. Suppose, for example, a random choice model P on the universe of the "Latitude" domain satisfies the multiplicative inequality. Then the probability of choosing Warsaw from the menu of Warsaw, London and Vancouver must be at least as great as the joint probability of two independent events: choosing Warsaw over London in pairwise choice and then choosing Warsaw again in pairwise choice, this time over Vancouver. In the experiment, 17 out of 40 participants chose Warsaw from the tripleton menu, 22 out of 40 chose Warsaw over London and 31 out of 40 chose Warsaw over Vancouver. Since $0.425 = \hat{P}_{\{a,b,c\}}(a) < \hat{P}_{\{a,b\}}(a)\hat{P}_{\{a,c\}}(a) = 0.42625$, the system of choice *proportions* $\hat{P}$ violates the sample analogue of $P_{\{a,b,c\}}(a) \geq P_{\{a,b\}}(a)P_{\{a,c\}}(a)$, one of the multiplicative inequalities. A formal test of the multiplicative inequality (a condition on choice *probabilities*) will take into account sampling variation and the full set of inequalities. We perform such tests in Section 5.2 below.

Formally, a random choice model P satisfies the *multiplicative inequality* if and only if for all $x \in A, B \subseteq U$,

$$P_{A \cup B}(x) \geq P_A(x)P_B(x). \tag{3}$$

Sattath & Tversky (1976) show that it is a testable implication of the independent random utility model and also of the Elimination by Aspects (EBA) model described in Tversky (1972*a*) and Tversky (1972*b*). McCausland & Marley (2024) show that any given configuration of choice probabilities on the binary menus of a universe can be extended to a random choice model (by specifying choice probabilities on the remaining, larger, menus) in such a way that the random choice model satisfies the multiplicative inequality. As with regularity, each P_A is pivotal.

The conditions we consider here are just three examples among the many that have been studied.

Fishburn (1999) documents these and many other conditions on choice probabilities, including ones that have been proposed as reasonable assumptions about choice behaviour and some that are testable implications of different classes of random choice models. For example, the paper lists nine conditions on binary choice probabilities that are "motivated in part" by the transitivity axiom on preferences that underlies most of economic theory. Another condition on binary choice probabilities, the triangle inequality, is a necessary condition for regularity and is therefore implied by any random utility model. A further example of a condition that jointly constrains choice probabilities across all menus is order-independence, which is necessary and sufficient for models satisfying simple scalability. See Luce & Suppes (1965) and McCausland & Marley (2013) for illustrations of the logical relationships among a set of conditions that includes the three outlined here and various kinds of stochastic transitivity.

2.3 Individual- and population-level choice

Most experiments involve a sequence of choice trials, where some menu is presented at each trial. We distinguish two different types of repeated choice behaviour. We will qualify choice behaviour as *individual*-level when it consists of the repeated choices of a single individual and *population*-level when a participant is chosen at random from a population at each trial. Hybrid experiments are possible, but we are not concerned with these here. We have seen that our experiment has a population-level design.

A random choice model can be used to describe either individual- or population-level behaviour, but we should not expect the same kinds of model to apply to both. For example, we would expect more heterogeneity in population-level choice, and thus fewer choice probabilities close to zero or one. Whether or not a condition like regularity applies to individual-level choice and whether or not it applies to population-level choice may both be of interest, but they are two distinct scientific questions.

That said, the two questions are sometimes related, through the following aggregation property: if the random choice model for all individuals in a population satisfies the condition, the model describing the population also satisfies the condition. Expressed in contrapositive form to make plain the implications for testing, if the condition is violated for a population, then it must be violated by at least one individual.

Some, but not all, conditions have this aggregation property. Fortunately, a simple mathematical test tells us whether or not any given condition does or does not. Define Δ as the set of all

random choice models and Λ as the subset of models satisfying a given condition of interest. Then the condition satisfies the aggregation property if and only if Λ is a convex set. A set is convex if every non-negatively weighted average of points in the set also belongs to the set. Here, the points represent random choice models, and taking a weighted average corresponds to combining the choice probabilities of different individuals into population choice probabilities. If all individual random choice models lie in the same convex set, then the population random choice model must also. Conversely, if the region where the condition holds is not convex, we can find two individual random choice models that satisfy the condition such that the random choice model for the population of two does not.

Regularity and random utility satisfy the aggregation property because the regions in which they hold are convex.[1] McCausland & Marley (2024) show that the region in which the multiplicative inequality holds is not convex. Therefore it does not satisfy the aggregation property.

Another issue concerning individual- and population-level choice is the assumption, which we maintain throughout, that observed choices are statistically independent and for each menu, identically distributed. There are good reasons to be sceptical of this assumption for individual-level choice. Participants may be learning or their attention may be waning during the course of an experiment. The hypothesis is more plausible for population-level choice, where variation across observations comes from random sampling and not from changes in the behaviour of any individual.

2.4 Context effects

Context effects are examples of choice behaviour that are regarded as anomalous because they violate stochastic choice conditions that are either considered plausible or implied by important classes of discrete choice models. To provide challenging environments for stochastic choice conditions, we deliberately design twelve of the choice domains to be conducive to one or more of three well-documented context effects: the attraction, similarity and compromise effects. Here we review the effects and give some examples. In Appendix B, where we describe all of the choice domains in detail, we describe how those twelve domains are designed to be conducive to these effects. Section 5.1 gives an empirical illustration where we test directly for an attraction effect in one of the domains.

The three context effects can be seen as contingent conditions on choice probabilities, conditions

[1] The regularity inequalities are all linear in choice probabilities, and so the regularity region is the intersection of the convex set Δ and many half-spaces, also convex. Likewise, the Block-Marschak inequalities are also linear in choice probabilities and so the random utility region is convex.

predicted to hold when the items in question are in certain relationships with each other. The example of the attraction effect in the introduction pertains to the items Vancouver, Seattle and Paris. The condition predicted to hold is that the probability of choosing Vancouver goes up when Seattle is added to the menu of Vancouver and Paris. The prediction is contingent on Seattle being clearly "inferior to" (i.e. obviously further south than) Vancouver. We will describe seminal versions of each of the three effects. Other versions of these context effects have been proposed, and they have different aggregation properties and different implications on whether choice probabilities can satisfy regularity, the random utility condition, or other conditions not considered here, such as Krantz's (1967) simple scalability. See Davis-Stober et al. (2023) for a comprehensive discussion.

The attraction effect (sometimes known as the asymmetric dominance effect) was introduced by Huber et al. (1982), and pertains to choice environments with three items and some dominance relation over these items. In one experimental design from Huber et al. (1982), items are six-packs of beer with two attributes, price and quality rating. A six-pack dominates another when its price is lower and its quality higher at the same time. The three items are a "target" x, a "competitor" y and a "decoy" z; x dominates z, but y does not; neither x nor y dominates the other. In the six-pack example, the target x has a price of \$1.80 and a rating of 50. The competitor y has a higher price (\$2.60) but also a higher rating (70) than the target, so neither dominates the other. The decoy z has a higher price (\$2.20) than the target and a lower rating (40). The target dominates the decoy because it is cheaper and has a higher rating. Importantly, the competitor does not dominate the decoy; this establishes an asymmetry between the target and the competitor. The effect is said to occur when the probability of choosing the target goes up when the decoy is added to the pair of target and competitor; that is, when $P_{\{x,y,z\}}(x) > P_{\{x,y\}}(x)$. This condition is inconsistent with regularity, since one of the regularity inequalities is $P_{\{x,y\}}(x) \geq P_{\{x,y,z\}}(x)$; and the random utility condition, because regularity is a necessary condition for random utility. The "Beer" domain of our experiment is based on this design, but includes five items (and prices updated from 1982).

We also include domains where the dominance relation is not based on explicit levels of attributes. One example is from the "Latitude" domain, a configuration with Vancouver as target, Warsaw as competitor and Seattle as decoy. Two other configurations feature London and Paris, respectively, as competitor in place of Warsaw.

Tversky (1972*b*) introduced the similarity effect, also pertaining to choice environments with three items. Here, there is a similarity relation among the choice items under consideration. Our "Dots" domain is similar to one of the three experimental designs in Tversky (1972*b*) and includes

boxes with 320, 310, 300, 290 and 280 points, labelled a, b, c, d and e, respectively. The prompt question is "Which one of the following boxes do you think has the greatest number of points?" We can define a similarity relation that considers two adjacent boxes in this list as similar. The similarity effect occurs if there are items x, y and z such that x and y are similar, x and z are not similar and y and z are not similar and choice probabilities satisfy

$$P_{\{x,z\}}(x) > \frac{P_{\{x,y,z\}}(x)}{P_{\{x,y,z\}}(x) + P_{\{x,y,z\}}(z)} \quad \text{and} \quad P_{\{y,z\}}(y) > \frac{P_{\{x,y,z\}}(y)}{P_{\{x,y,z\}}(y) + P_{\{x,y,z\}}(z)}. \tag{4}$$

The inequality on the left holds if adding the option y, similar to the target option x, to the menu $\{x, z\}$ lowers the *relative* probability of choosing x, compared to z. The inequality on the right is similar, except that y is the target option and x is the similar option. The tripleton set $\{a, b, e\}$ is a candidate for the similarity effect, with a and b as the similar items and e as the dissimilar item. Likewise, $\{a, d, e\}$ is also a candidate, with d and e as the similar items.

Simonson (1989) introduces a strong form of the compromise effect and Tversky & Simonson (1993) introduce a weaker version. They pertain to choice environments where items can be described in terms of a set of real-valued attributes. An item y is *between* items x and z if two conditions hold: first, that for each attribute, y has a level of that attribute intermediate between the levels of x and z; and second, that each of x and z has the highest level of at least one attribute. The weak effect is said to occur when y is between x and z and the following inequalities, from page 1183 in Tversky & Simonson (1993), hold:

$$P_{\{y,x\}}(y) < \frac{P_{\{x,y,z\}}(y)}{P_{\{x,y,z\}}(y) + P_{\{x,y,z\}}(x)} \quad \text{and} \quad P_{\{y,z\}}(y) < \frac{P_{\{x,y,z\}}(y)}{P_{\{x,y,z\}}(y) + P_{\{x,y,z\}}(z)}. \tag{5}$$

The inequality on the left holds if adding the option z to the menu $\{x, y\}$, making y a "compromise" between x and z, raises the relative probability of choosing y, compared to x. The inequality on the right is similar, except that x is added to $\{y, z\}$, again making y a "compromise" between x and z. Our "Beer" domain includes three six-packs where a betweenness relation, relative to the price and quality rating attributes, applies.

The similarity and compromise effects are compatible with the conditions we consider in this paper, but not with others that may be of interest, such as the constant ratio rule described by Tversky (1972*a*). See Davis-Stober et al. (2023) for details; different versions of context effects may differ in terms of their compatibility with different conditions.

3 Experimental design

Here we describe the experiment, pointing out design features that serve the objective of testing abstract stochastic choice conditions across a wide variety of choice environments.

3.1 Selection of choice domains

An important objective is to cover a wide variety of choice environments. For this reason we chose domains to span several qualitatively different kinds of choice, including consumer choice, aesthetic preference, factual judgment and perceptual discrimination. For example, the "Beer" domain mentioned above mimics domains used in consumer choice experiments where different items have different levels of various attributes. In the "Juice" domain, participants choose among different kinds of juice. In some domains, participants make judgments, such as the "Sentences" domain, where they make relative judgments about grammaticality. In some domains, like the "Dots" domain described above, choices involve perception. Our experiment features 32 choice domains in total, each domain consisting of a universe of five related choice items.

Table 2 shows the names of the choice domains, together with the prompt (or an excerpt of the prompt, with excluded text indicated by an ellipsis) used to elicit a response from the participant. Appendix B describes each domain in more detail, giving the full prompt, describing the five choice items and, where applicable, providing a source for the information used to construct the question. Figure 1 shows screenshots for the "Travel" and "Coffee" domains. The screenshot for the "Travel" domain shows a menu with two of the five items in that domain. The screenshot for the "Coffee" domain shows a menu with four of the five items in that domain.

To better stress-test stochastic choice conditions, we include many domains intended to be conducive to anomalous behaviour, in the form of context effects. Some examples we have seen above are the "Latitude", "Beer", and "Dots" domains. Appendix B describes these and several others in detail.

3.2 A population-level full-menu design

Many of the stochastic choice conditions one might want to test have the property that for each menu A, the probability vector P_A is pivotal, in the sense defined in Section 2. This is true for the three conditions we have considered in this paper. Because observed choices from any menu have the potential to provide evidence against these conditions, it is important to observe choices from

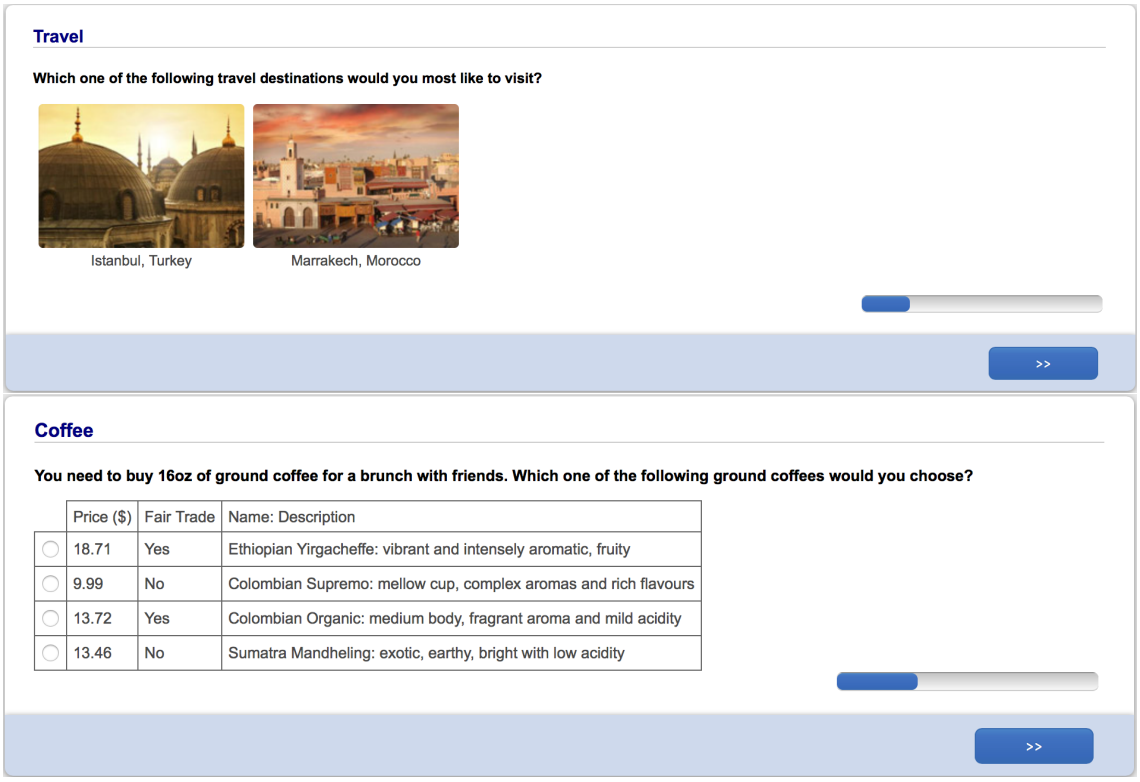


Figure 1: A screenshot for the "Travel" domain, in which participants select pictures, and a screenshot for the "Coffee" domain, in which participants select radio buttons. In the experiment and in electronic versions of this paper, these photographs are in colour.

Table 2: Choice domains of the experiment, with prompt questions. An ellipsis indicates missing text and is never part of a prompt. Appendix B gives the full text of all prompts.

Domain name	Question
Male stars	Which movie star would you choose to have lunch with?
Female stars	Which movie star would you choose to have lunch with?
Films	Judging from the following descriptions of films, which one of the films would you choose to see?
Star pairs	Knowing only who is starring, which one of these new films would you choose to see?
Pizzas	Which one of the following pizzas would you choose?
Juices	Which one of the following fresh juices would you choose?
Colours	Which one of the following colours do you like best?
Colour combos	Which one of these colour combinations do you like best?
Events	Which one of the following events do you think is most likely to happen in the next twenty years?
Radio formats	Suppose you were on a two hour road trip and you have a choice among radio stations with the following formats. Which one would you choose?
Music	Which one of the following musical artists do you like the best?
Aboriginal	Which one of the following examples of Australian aboriginal art do you like the best?
Impressionist	Which one of the following examples of Impressionist art do you like the best?
Sentences	Which one of the following sentences do you find the most grammatically acceptable?
Travel	Which one of the following travel destinations would you most like to visit?
Marijuana	Which one of the following marijuana policies would you choose?
Latitude	Which one of the following cities do you think is furthest north?
Dots	Which one of the following boxes do you think has the greatest number of points?
Triangles	Which one of the following triangles do you think has the greatest area?
Population	Which one of the following countries do you think has the largest population?
Surface area	Which one of the following countries do you think has the greatest surface area, including inland bodies of water?
Beer	... Given that you had to choose one brand to buy on this information alone, which one would you choose?
Cars	Which one of the following similarly priced cars would you choose to drive, based on the percentage of owners who indicated that they "would buy again," and the fuel economy rating?
Restaurants	Which one of the following restaurants would you choose for your next restaurant meal...?
Layovers	Which one of the following flight itineraries would you choose? ...
Future	Which one of the following would you choose?
Phone plans	Of the following cell phone plans, which one would you choose? ...
Hotel rooms	... Which one of the following hotels would you choose, ...?
Itineraries	Which one of the following flight itineraries would you choose? ...
Televisions	Which one of the following televisions would you choose to buy ...?
Coffee	Which one of the following ground coffees would you choose?
Charity	Suppose someone was donating a total of 100 dollars to a combination of charities, on your behalf. Which one of the following divisions of the 100 dollars would you choose?

every menu of a domain. We have settled on a purely population-level design, to ensure that for each domain, choices are plausibly independent and, menu by menu, identically distributed. For

these reasons, each participant sees exactly one menu from each domain, exactly once; and each menu of each domain is seen by several participants.

Here are the details of the design. There are $J = 32$ choice domains, each with a 5-item universe. For each domain, there are $I = 26$ menus (binary and larger subsets of the five-item universe), for a total of $JI = 832$ menus. Since each participant faces exactly one menu from each domain, once, the number of choice trials for each participant is the same as the number of domains, $J = 32$.

Participant assignments were generated in blocks of 26. Within each block and domain, the 26 participants were randomly assigned one-to-one to the 26 menus, with assignments made independently across blocks and domains. The intended design called for 40 complete blocks, yielding $40 \times 26 = 1040$ participants and exactly 40 observations per menu. However, two participants from the next block began the survey before the target had been reached and completed it afterward. Their choices were retained for analysis. Consequently, in each domain, two menus were presented 41 times and the remaining 24 menus were presented 40 times. Because every participant faced exactly one menu in each of the $J = 32$ domains, the realized dataset contains $1042 \times 32 = 33,344$ trials.

3.3 Other design features

Participants are constrained to choose exactly one option from each menu. For some domains, participants select radio buttons and in others, they select pictures, which are then identified by a border. Figure 1 shows one screenshot for each kind of domain. In each case, participants can modify their choice before they confirm it by clicking on a button labelled ">>"; once they click this button, they cannot go back and change their response. A progress bar indicates progress through the experiment.

Assignments of participants to menus, the order of trials and the positions of menu items on the screen were randomized. Menu assignments followed the block randomization procedure described in Section 3.2 and were mutually independent across blocks and domains. Elements of a menu were presented in random order (left to right or top to bottom, as the case may be) on the screen, independently across participants facing that same menu. The $J = 32$ trials assigned to each participant were presented in random order, independently across participants. Random assignments, positions and trial order were mutually independent.

3.4 The sample of participants

The experiment consisted of an internet survey conducted between August 10, 2017 and August 31, 2017. As explained in Section 3.2, 1042 participants completed the survey and are all included in the analysis. The experiment was approved by the Human Research Ethics Board at the University of Victoria and all participants gave informed consent before completing the survey. Participants were recruited by Survey Sampling International (SSI) from their Canadian panel, designed to be representative of the Canadian adult population. SSI screens potential members for reliability and requires that all participants be at least 18 years of age. It also verifies the identity of their participants for each survey. Compensation of participants includes cash and other rewards. Programming and hosting of the survey were provided by the Institute for Choice at the University of South Australia.

We chose not to impose any additional restrictions on our sample of participants. We have full information on age, province or territory, and sex for all 1042 participants in our sample. There are 515 men (49.4%) and 527 women (50.6%). Age ranges from 18 to 88; the percentage of participants in the age bracket 18-24 is 11.4%; 25-34, 17.8%; 35-54, 38.3%; 55 and older, 32.5%. Participants could take as long as they liked to complete the survey. The quartiles of total duration were 7.1, 10.0 and 13.9 minutes. Three participants took longer than 24 hours, and the shortest duration was 1.7 minutes. Participants were given the opportunity to leave written feedback on the experiment upon completion.

4 Bayesian models and inference

In Section 2, we defined a random choice model P on a universe U of items. The assumption that choices are independent, and for each menu identically distributed, completes the description of the data generating process for data structured like ours. We will denote by N the complete set of experimental data as a random vector, before observation. This includes choice counts from every item of every menu. We will denote by n a generic possible value of the data vector. We denote by n_{obs} the realized data vector we observe, such as the set of counts displayed in columns two through six of Table 1. The data generating process gives the conditional distribution of N given P.

We use Bayesian inference to learn about plausible values of the unknown P after observing $N = n_{\text{obs}}$. Because the goal is to test stochastic choice conditions, we conduct inference directly on

the space of all random choice models and not a reduced-dimension parametric class. That said, because there are a finite number of menus and options, P is a finite dimensional parameter vector. For a universe of size five, there are 75 choice probabilities subject to 26 adding-up constraints, leaving 49 free parameters. The parameter space, which we denote by Δ, is the set of all random choice models; it consists of all values of the vector P consistent with the axioms of probability.

Bayesian inference requires a prior distribution for P. Once we have this, we have a joint distribution of P and N and we can derive, in principle, the posterior distribution of P (i.e. the conditional distribution of P given N) and the marginal distribution of N. We will call the unknown P, completed by a prior distribution over P, a *Bayesian random choice model*. The use of the word "model" here is consistent with its use in the term "random choice model". Just as a particular known random choice model describes an unconditional distribution for N, a Bayesian random choice model implies an unconditional distribution for N.

Our main inferential objective is testing stochastic choice conditions using data, which in a Bayesian context is an exercise in model comparison. To test a given condition, we will compare two Bayesian random choice models, an *encompassing* model where no restrictions are placed on P beyond the axioms of probability, with a *constrained* model, where P is restricted to the set of random choice models that satisfy the condition. To do this we will use two different kinds of prior distribution. The encompassing model is a Bayesian random choice model where the prior distribution of P does not rule out any random choice models *a priori*; that is, the prior has full support on Δ. The constrained model is a Bayesian random choice model where the prior distribution is the truncation of the prior distribution of the encompassing model to the region where the condition holds. Throughout the paper, Λ denotes the subset of random choice models satisfying the condition under consideration; each tested hypothesis has the form $P \in \Lambda$.

The standard summary statistic for a comparison of two Bayesian models is the Bayes factor. If two Bayesian random choice models, M and M', are both considered equally likely before seeing the data, the Bayes factor in favour of M' over M gives the posterior odds ratio between them, $\Pr[M'|N = n_{\text{obs}}]/\Pr[M|N = n_{\text{obs}}]$. So for example, if the two models are equally likely to begin with and the Bayes factor is 2, the posterior probability assigned to model M' is twice as high as that assigned to M.

The Bayes factor in favour of M' over M is defined as a ratio of marginal likelihoods:

$$\text{BF} \equiv \frac{f_{M'}(n_{\text{obs}})}{f_M(n_{\text{obs}})}, \tag{6}$$

where $f_M(n)$ is the marginal probability mass function of the data vector under model M, and $f_{M'}(n)$ is the same under M'. Evaluating $f_M(\cdot)$ at the observed data n_{obs} gives a measure of the prior predictive performance of M for the observed data. Because it averages over all parameter values rather than evaluating a model's predictive performance at a fitted estimate, it measures predictive performance on unseen data.

For the special case when M is an encompassing model and M' is a constrained model obtained by truncating the prior distribution of M to the region Λ of Δ where a condition of interest holds, the Bayes factor reduces to the following intuitively satisfying form:

$$\text{BF} = \frac{\Pr_M[P \in \Lambda | N = n_{\text{obs}}]}{\Pr_M[P \in \Lambda]}. \tag{7}$$

The numerator is the posterior probability, under the encompassing model M, of the hypothesis $P \in \Lambda$, the hypothesis that the condition holds. As such, the numerator measures the compatibility of the condition with the observed data n_{obs}. The denominator, the prior probability of the same hypothesis under the encompassing model, is an inverse measure of model parsimony; the lower the denominator, the more restrictive the condition and the more parsimonious the constrained model. See Klugkist & Hoijtink (2007) for details. For further explanation of Bayes factors, see Berger (1985) or Bernardo & Smith (1994).

The ratio in (7) also suggests a practical way of computing numerical estimates of Bayes factors. Since the probability of an event is just the expectation of an indicator function for the event, we can estimate numerator and denominator using Monte Carlo methods for estimating expectations. In the next section we turn to some empirical analyses, where we report Bayes factors estimated numerically in this way in order to compare various Bayesian random choice models.

5 Empirical Illustrations

We begin, in Section 5.1, with a simple context effect analysis involving the comparison of just two choice probabilities. This illustrates the Bayesian testing framework in a simple setting. Then, in Section 5.2, we perform global tests of the three stochastic choice conditions outlined in Section 2. We end, in Section 5.3, by showing a graphical representation of all binary and ternary choice proportions for a domain, a representation that can reveal potential violations of regularity and the multiplicative inequality.

Target	Competitor	Decoy	$(n_T^{(2)}, n_C^{(2)})$	$(n_T^{(3)}, n_C^{(3)}, n_D^{(3)})$	$\Pr[P \in \Lambda \mid N = n_{\text{obs}}]$	BF	$\ln \text{BF}$
Vancouver	Warsaw	Seattle	$(9, 31)$	$(9, 30, 2)$	0.453	1.359	0.306
Vancouver	London	Seattle	$(11, 29)$	$(15, 22, 3)$	0.805	2.415	0.882
Vancouver	Paris	Seattle	$(17, 23)$	$(26, 14, 0)$	0.969	2.907	1.067
London	Warsaw	Paris	$(18, 22)$	$(14, 24, 2)$	0.162	0.486	-0.722
London	Vancouver	Paris	$(29, 11)$	$(24, 14, 2)$	0.096	0.289	-1.240
London	Seattle	Paris	$(31, 9)$	$(25, 7, 8)$	0.056	0.169	-1.780

Table 3: Data and test results for six candidate attraction effects in the Latitude domain. Each row gives a configuration of target, competitor and decoy options where we might expect an attraction effect. The fourth column gives the numbers of participants choosing the target and competitor when the menu is this pair of options. The fifth column gives the numbers of participants choosing the target, competitor and decoy when the menu is the set of these three options. The inconsistent but compact notation in the 4th and 5th column names is used only here. The sixth column gives the posterior probability of the hypothesis that the attraction effect holds. The last two columns give the Bayes factor and (natural) log Bayes factor in favour of the hypothesis that the attraction effect holds.

5.1 Tests of context effects using Bayes factors

Here we provide a worked example showing how to test the hypothesis that a context effect holds in a given situation. We consider six possible configurations of target, competitor and decoy in the "Latitude" domain, based on the dominance relation that puts Vancouver over Seattle, as proposed in Section 1, and London over Paris. Each row of Table 3 describes one configuration, provides the relevant choice counts and reports test results for the hypothesis that the effect holds. The first three columns identify the target (T), competitor (C) and decoy (D) options that define the configuration. The fourth column gives $n_{\{T,C\}}(T)$ and $n_{\{T,C\}}(C)$, the number of participants choosing the target and competitor, respectively, from the menu of these two options. The fifth column gives $n_{\{T,C,D\}}(T)$, $n_{\{T,C,D\}}(C)$ and $n_{\{T,C,D\}}(D)$, the number of participants choosing the target, competitor and decoy, respectively, from the menu of these three options. We see that the number of people choosing Vancouver goes up in the presence of Seattle, the decoy, when the competitor is either London or Paris. This is suggestive evidence for two attraction effects.

Observations from menus consisting of the proposed target and decoy options provide some independent evidence on the plausibility of the two dominance relations, and these suggest that Seattle is a more convincing decoy for Vancouver than Paris is for London. Of the participants who saw the menu of Vancouver and Seattle, 36 chose Vancouver and 4 chose Seattle. Of those who saw the menu of London and Paris, only 32 chose London and 8 chose Paris.

For each configuration, the hypothesis is that the probability of choosing the target increases when the decoy is added to the menu. The analysis therefore reduces to inference about two unknown probabilities and whether one exceeds the other. We use an encompassing model with a prior distribution where all probability vectors are independent and uniformly distributed. In particular, the probability vectors of interest, $P_{\{T,C\}} = (P_{\{T,C\}}(T), P_{\{T,C\}}(C))$ and $P_{\{T,C,D\}} = (P_{\{T,C,D\}}(T), P_{\{T,C,D\}}(C), P_{\{T,C,D\}}(D))$ are independent and each is uniformly distributed on the region where its elements are non-negative and add to one. The hypothesis we wish to test is that the attraction effect holds, or equivalently, that $P_{\{T,C,D\}}(T) > P_{\{T,C\}}(T)$. Taking Λ to be the region of Δ where the condition $P_{\{T,C,D\}}(T) > P_{\{T,C\}}(T)$ holds, the constrained model has a prior distribution for P that is the truncation of the prior distribution in the encompassing model to the region Λ.

We use standard results for the Dirichlet-multinomial and beta-binomial models (see for example Gelman et al. (1995)) to analytically derive the posterior distribution of $P_{\{T,C\}}$ and $P_{\{T,C,D\}}$ under the encompassing model. In the following derivation, we denote by $\mathrm{Be}(\alpha, \beta)$ the beta distribution with shape parameters α and β. The prior independence of $P_{\{T,C\}}$ and $P_{\{T,C,D\}}$ and the assumed independence of participants' choices imply the posterior independence of $P_{\{T,C\}}$ and $P_{\{T,C,D\}}$. The uniform distributions of $P_{\{T,C\}}$ and $P_{\{T,C,D\}}$ imply that the prior distribution of $P_{\{T,C\}}(T)$ is $\mathrm{Be}(1,1)$ and that that of $P_{\{T,C,D\}}(T)$ is $\mathrm{Be}(1,2)$. Then the posterior distribution of $P_{\{T,C\}}(T)$ is $\mathrm{Be}(1 + n_{\{T,C\}}(T), 1 + n_{\{T,C\}}(C))$ and that of $P_{\{T,C,D\}}(T)$ is $\mathrm{Be}(1 + n_{\{T,C,D\}}(T), 2 + n_{\{T,C,D\}}(C) + n_{\{T,C,D\}}(D))$.

The test for the attraction effect is based on the Bayes factor in favour of the constrained model, where the condition $P_{\{T,C,D\}}(T) > P_{\{T,C\}}(T)$ holds, over the encompassing model, where there are no restrictions on any choice probabilities. This is computed as $\Pr[P \in \Lambda | N = n_{\mathrm{obs}}] / \Pr[P \in \Lambda]$, the ratio of posterior to prior probability of the effect under the encompassing model. We computed $\Pr[P \in \Lambda] = \frac{1}{3}$ analytically. We estimated $\Pr[P \in \Lambda | N = n_{\mathrm{obs}}]$ numerically with Monte Carlo simulation methods, using Rao-Blackwellization to reduce variance. We computed numerical standard errors, which measure the standard deviation of simulation error.

Column six of Table 3 gives numerical estimates of $\Pr[P \in \Lambda | N = n_{\mathrm{obs}}]$, the posterior probability of the attraction effect under the encompassing model. All numerical standard errors are less than 0.0001. The final two columns give the Bayes factor (BF), equal to the ratio $\Pr[P \in \Lambda | N = n_{\mathrm{obs}}] / \Pr[P \in \Lambda]$, and its natural logarithm.

The evidence favours the attraction effect for the three cases where the target is Vancouver and

the decoy is Seattle. In contrast, it favours the encompassing model for the three cases where the target is London and the decoy is Paris.

We can use the guidelines recommended by Kass & Raftery (1995) to classify the strength of the evidence provided by the Bayes factors. They classify log Bayes factors[2] between 0 and 1 as "not worth more than a bare mention", those between 1 and 3 as providing "positive evidence", those between 3 and 5 as providing "strong evidence" and those above 5 as providing "very strong evidence". We will adopt these categories throughout the paper, but refer to the first two as giving "weak" and "moderate" evidence.

By this reckoning, there is moderate evidence in favour of the attraction effect when the target is Vancouver, the competitor is Paris and the decoy is Seattle and moderate evidence against the attraction effect when the target is London, the decoy is Paris and the competitor is either Vancouver or Seattle.

We can also consider joint hypotheses, such as the hypothesis that the attraction effect holds in all three cases where the target is Vancouver and the decoy is Seattle. This joint hypothesis consists of three inequalities; the log Bayes factor in favour of the hypothesis is the sum of the individual log Bayes factors, or 2.253. This is stronger evidence, but still not strong according to the standards of Kass & Raftery (1995). The log Bayes factor in favour of the joint hypothesis that the attraction effect holds in all three cases where the target is London and the decoy is Paris is -3.744, which implies strong evidence against this joint hypothesis.

5.2 Tests of three global abstract stochastic choice conditions

Regularity, the random utility condition and the multiplicative inequality are also joint hypotheses. However, the number of inequalities involved is much greater. Regularity, for example, consists of 70 distinct and non-redundant inequalities when the universe consists of five items.

It is natural to choose a prior distribution where the P_A are mutually independent across menus. We did this above when we tested for multiple attraction effects. Cavagnaro & Davis-Stober (2014), Davis-Stober et al. (2015), Myung et al. (2005) and Zwilling et al. (2011) do the same for testing simpler and less restrictive conditions than the three we investigate here.

Although the simplicity of such a prior is in its favour, some kinds of prior correlation between choice probabilities on different menus are intuitively compelling. Suppose you observe choices only from the menu $\{a, b, c\}$. You don't know what the options a, b and c are but see that many

[2] All logarithms are natural (base e) logarithms.

more people choose a than b. Now you appraise choice probabilities on the smaller set $\{a, b\}$. We suspect most people would put more weight on higher values of the probability $P_{\{a,b\}}(a)$ than they would before observing the choices from $\{a, b, c\}$; this, despite not knowing what the options are and having no direct evidence from observing choices from $\{a, b\}$. This is consistent with Bayesian reasoning, but not when $P_{\{a,b\}}$ and $P_{\{a,b,c\}}$ are deemed *a priori* independent.

Furthermore, computational problems arise when we test our three conditions using a prior where the P_A are independent. For a universe of five items, the prior probabilities of these conditions are so tiny that the amount of simulation required to measure them using straightforward simulation would be prohibitively high. The low prior probability of these conditions is a measure of how restrictive they are, compared with, for example, conditions on binary choice probabilities studied in the work cited above.

McCausland & Marley (2013) address both these issues by introducing a parametric prior distribution for P where the P_A are dependent. For testing abstract stochastic choice conditions, the prior has some attractive properties, such as invariance to relabelling choice items. Following McCausland & Marley (2014), we used this class of priors for our empirical analysis. In order to describe a prior sensitivity analysis we do below and to explain a difference between our analysis here and that in McCausland & Marley (2014), we digress to describe the parameters of this prior distribution.

The prior has two parameters, $\lambda \in [0, 1]$ and $\alpha > 0$. The λ parameter controls the degree of dependence of choice probabilities across menus. When $\lambda = 0$, the P_A are mutually independent. Thus, if you adopt a prior where $\lambda = 0$, you learn nothing about $P_{\{a,b\}}(a)$, even if you observe a thousand people out of a thousand choosing a from $\{a, b, c\}$. As λ increases, the prior probability of regularity and the random utility condition both increase. The marginal distribution of each P_A does not depend on λ, but correlations of choice probabilities across menus do. For example, the correlation between $P_{\{a,b,c\}}(a)$ and $P_{\{a,b\}}(a)$ increases in λ, even though their two marginal distributions do not depend on λ. The prior is constructed so that at the maximal value $\lambda = 1$, the probabilities of regularity and the random utility condition are both one.

The α parameter governs the degree to which choice probabilities concentrate near zero or one, with lower values of α corresponding to more probability mass near zero or one. Thus, low values of α are associated with more extreme choice probabilities and high values of α with less extreme choice probabilities. While we use the same family of priors here that we did for previous analyses in McCausland & Marley (2014) and McCausland et al. (2020), some details are different. The

main reason is that the previous papers study individual-level choice and the current paper studies population-level choice. We expect heterogeneity across individuals. Therefore we anticipate more choice variability and thus less extreme choice probabilities at the population level. For this reason, we favour higher values of α and therefore less prior probability mass for choice probabilities near zero and one, compared with those previous papers.

As in these previous papers, we conducted a prior sensitivity analysis. Here, we tested each stochastic choice condition using three different encompassing models, M_1, M_2 and M_3, which differ only in terms of the prior distribution of λ. M_1 favours lower values of λ than M_2 and M_3 favours higher values. The priors were chosen to represent progressively stronger prior belief in dependence of choice probabilities across menus. The prior probability of the random utility condition is 0.002 under M_1, 0.051 under M_2 and 0.228 under M_3. We provide details of the three prior specifications in Appendix A.

We used the posterior simulation and model comparison methods described in McCausland & Marley (2014) for the computation of all results reported in Section 5.2. For each choice domain, we computed log Bayes factors in favour of the encompassing models M_1 and M_2, over the encompassing model M_3. For each encompassing model, we computed log Bayes factors in favour of three constrained models, over the encompassing model. Those constrained models are the ones obtained by truncating the prior density for the encompassing model to the regions where regularity, random utility and the multiplicative inequality, respectively, hold.

Recall that log marginal likelihoods measure a model's prior predictive performance. Log Bayes factors, which compare prior predictive performance between two models, are the differences of their log marginal likelihoods. Therefore we can compute the log Bayes factor of any of the twelve models over any other as a difference between two of the log Bayes factors mentioned above.

We report log Bayes factors for the twelve models, all relative to the encompassing model M_3. The choice of M_3 as a reference model, which is the most favoured encompassing model for 31 out of the 32 domains, just serves to establish a level of zero on a scale of prior predictive performance; by definition, the log Bayes factor for M_3 relative to itself is zero.

Figure 2 gives, for each choice domain, log Bayes factors for various combinations of condition and encompassing model, relative to encompassing model M_3. There is a panel for each domain, and the vertical scale in each panel measures the log Bayes factor of models relative to M_3. Each symbol represents a model; the higher the symbol, the greater the log Bayes factor relative to M_3, and the better the model's prior predictive performance. Symbols in column i, $i = 1, 2, 3$, represent models

based on the encompassing model M_i. Circles represent the encompassing models themselves; triangles, squares and crosses represent constrained models obtained by imposing regularity, the random utility condition and the multiplicative inequality, respectively.

Within a panel, log Bayes factors between any two models out of twelve can be read off as the vertical distances between two points. The most relevant log Bayes factors are those comparing a constrained model with its corresponding encompassing model. Each of these log Bayes factors summarizes the evidence for a stochastic choice condition, using a particular encompassing model. They are given by the vertical distance between a symbol and the circle in the same column. So for example, the vertical distance between the triangle and the circle in column 1 of the top left panel "Male stars" gives the evidence in favour of regularity, using the encompassing model M_1, for the Male stars domain.

Recall that variation in results among the three encompassing models is used for prior sensitivity analysis. Comparisons of Bayes factors across columns illustrate how results depend on changes to the prior specification.

There are no crosses for the choice domains "Latitude", "Layovers" and "Future", regardless of the encompassing model. This is because the posterior probability of the multiplicative inequality is too close to zero to measure, implying very strong evidence against that condition for those domains. Evidence in support of the multiplicative inequality varies a lot across choice domains, from moderate evidence in favour to very strong evidence against. In each case, evidence is fairly robust to the choice of encompassing model.

The evidence for regularity and the random utility condition tends to be weaker, but more consistent across choice domains. In most cases, the log Bayes factors either favour the random utility condition and regularity over the encompassing models, or are very close to zero, representing little evidence either way. Across domains and encompassing models, the evidence for the random utility condition is generally similar to that for regularity, which is implied by random utility. When the two differ, the evidence for regularity is usually slightly stronger, indicating some evidence against the higher order features of random utility, but this evidence is quite weak. For individual domains, the evidence in favour of regularity and the random utility condition are at best slight, but collectively, the evidence is stronger.

Figure 2 also allows us to compare two different ways of generating structure in choice probabilities: on the one hand, imposing a condition such as regularity or the random utility condition, and on the other, increasing prior dependence across menus through λ. We see that the variation

in predictive performance of the random utility condition across encompassing models (height of the square, across columns, for a particular domain) is much less than the variation in predictive performance across the encompassing models themselves (height of the circle, across columns, for the same domain). These results show that the two ways of generating structure in choice probabilities are somewhat complementary; some of the predictive gains from imposing the random utility condition can be replicated by increasing λ. The same observation applies to regularity.

5.3 Data visualization for observed choice proportions

The analyses above summarize evidence through Bayes factors. We conclude with a graphical representation of the observed choice proportions on all binary and ternary menus of the "Latitude" domain. While purely descriptive and not inferential, the representation can reveal patterns that suggest violations of regularity and the multiplicative inequality. Corresponding representations for all 32 domains are available in the supplementary materials.

We apply a graphical representation introduced by McCausland & Marley (2013) for plotting a random choice model, for a universe of size three, as a set of four points in a Barycentric coordinate system. The random choice model can always be the restriction to a universe of size three of a random choice model on a larger universe. Because systems of observed choice *proportions* satisfy the same non-negativity and adding-up constraints that random choice models do, the same graphical representation can be used to plot the observed choice proportions for all the menus of size two and three from the same set of three choice items.

Recall that for the "Latitude" domain, the universe of items is $\{a, b, c, d, e\}$ where a is Warsaw, b is London, c is Vancouver, d is Paris, and e is Seattle. Table 1 tabulates observed choice counts and choice proportions for all menus of this domain.

Figure 3 displays ten panels, one for each tripleton subset of $\{a, b, c, d, e\}$. The panel for a given tripleton shows an equilateral triangle with each vertex labelled with one of the items of that subset. Consider the single panel in the fourth row, for the tripleton $T \equiv \{c, d, e\}$, which we will use as an extended example. This triple corresponds to the configuration of Vancouver (c), Paris (d) and Seattle (e) where we saw the most pronounced evidence for an attraction effect. The vertices of the triangle are labelled c (bottom right), d (top) and e (bottom left). Four menus are subsets of T, $\{c, d\}$, $\{c, e\}$, $\{d, e\}$ and $\{c, d, e\}$; the choice proportions observed for these menus appear in rows 8, 15, 19 and 23 of Table 1, respectively.

The equilateral triangle in the fourth row is a Barycentric coordinate system; in it, we plot the

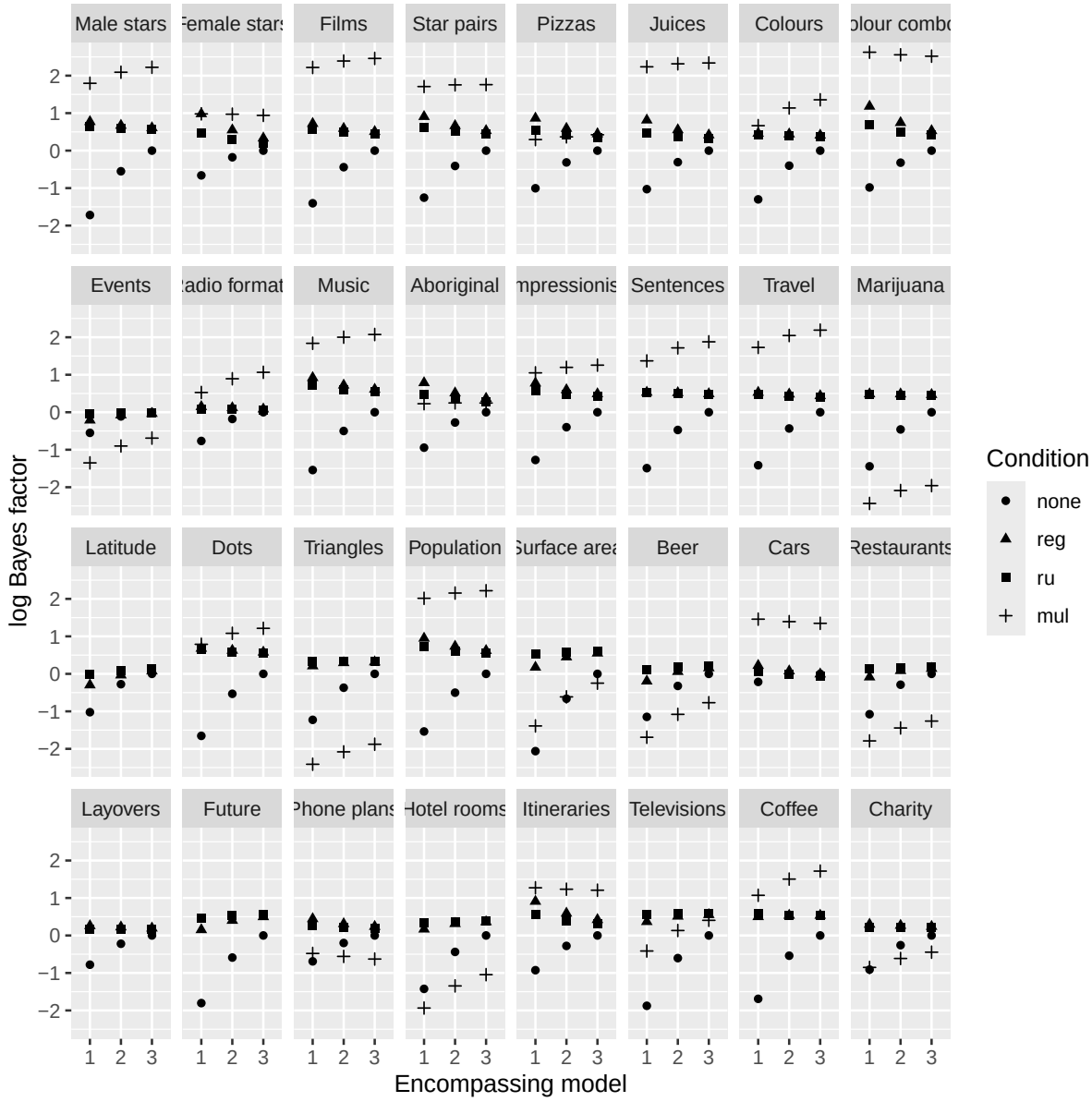


Figure 2: Log Bayes factors for 32 domains, by encompassing model and condition. The vertical scale gives log Bayes factors relative to the encompassing model M_3. The "Encompassing model" indices 1, 2 and 3 on the horizontal axes correspond to models M_1, M_2 and M_3. The circle (none) represents an encompassing model. The triangle (reg), square (ru) and cross (mul) represent constrained models, subject to regularity, the random utility condition and the multiplicative inequality, respectively. For any given domain, the vertical distance between a symbol and the circle in the same column is the log Bayes factor in favour of the corresponding condition for a given encompassing model.

observed choice proportions for these four menus. We represent the vector $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e)) = (\frac{26}{40}, \frac{14}{40}, \frac{0}{40}) = (0.65, 0.35, 0.00)$ of choice proportions observed in the experiment as a solid point. There are two ways of locating this point on the triangle. First, we can take vertices c, d and e as points on the plane of the page or screen. Then the point $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ is the convex combination of the vertices c, d and e—which have Barycentric coordinates $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, respectively—with weights 0.65, 0.35 and 0.00. Thus, the solid point is at $0.65c + 0.35d$. The second way relies on the fact that the triangle is equilateral. Taking the triangle to have unit height, the point representing $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ is at a distance of $\hat{P}_T(c) = 0.65$ to the left side of the triangle, at a distance of $\hat{P}_T(d) = 0.35$ to the base and at a distance of $\hat{P}_T(e) = 0.00$ to the right side. The upward sloping grid lines, the horizontal grid lines, and the downward sloping grid lines measure out the choice proportions of c, d and e, respectively.

The hollow dots on the right, left and bottom sides of the triangle represent the choice proportions $\hat{P}_{\{c,d\}}(c) = \frac{17}{40} = 0.425$, $\hat{P}_{\{d,e\}}(d) = \frac{26}{40} = 0.65$ and $\hat{P}_{\{c,e\}}(c) = \frac{36}{40} = 0.9$, respectively. For example, the hollow dot on the right side of the triangle is the convex combination of the vertices labelled c and d, with weights 0.425 and 0.575 respectively. Equivalently, the distance from this hollow dot to vertex c and the distance to vertex d are in the proportions 0.575 and 0.425.

The two triangles in the interior allow readers to detect, at a glance, if binary and ternary choice proportions for the tripleton set T violate regularity or the multiplicative inequality. [3] Suppose that we know the binary choice proportions $\hat{P}_{\{c,d\}}(c) = \frac{17}{40} = 0.425$, $\hat{P}_{\{d,e\}}(d) = \frac{26}{40} = 0.65$ and $\hat{P}_{\{c,e\}}(c) = \frac{36}{40} = 0.90$, but we have not yet observed the ternary choice proportions $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$.

The downward pointing solid triangle with the blue boundary is a cross-section of the region where regularity holds. For the three given binary choice proportions (indicated by the three hollow dots) the solid triangle with the blue boundary is the set of possible ternary choice proportions consistent with regularity and the three observed binary choice proportions. The observed ternary proportions (the solid dot) may or may not lie in the blue triangle; here they do not. The blue triangle is defined by the regularity inequalities on T, which simplify to $\hat{P}_T(c) \leq \min(\hat{P}_{\{c,d\}}(c), \hat{P}_{\{c,e\}}(c))$, $\hat{P}_T(d) \leq \min(\hat{P}_{\{c,d\}}(d), \hat{P}_{\{d,e\}}(d))$ and $\hat{P}_T(e) \leq \min(\hat{P}_{\{c,e\}}(e), \hat{P}_{\{d,e\}}(e))$. The observed choice proportion vector from menu T, $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$, lies outside the triangle with the blue boundary because $\hat{P}_T(c) = 0.65$ is greater than $\hat{P}_{\{c,d\}}(c) = 0.425$; we conclude that the binary and ternary

[3] Strictly speaking, regularity and the multiplicative inequality are restrictions on choice probabilities, not proportions. But a system of choice proportions has the same structure as a random choice model and belongs to the same space Δ. We define regularity and the multiplicative inequality for a system of choice *proportions* as the analogous conditions, with choice proportions replacing choice probabilities.

choice proportions on T do not satisfy regularity. It is important to remember that the observed choice proportions are subject to sampling variation and so the fact that choice proportions violate regularity does not constitute a rejection of the regularity hypothesis. We are describing here a data visualization method and not the results of statistical tests.

Similarly, the upward pointing solid triangle with the red boundary is a cross-section of the region where the multiplicative inequality holds. For the given binary choice proportions, this solid triangle is the set of ternary choice proportions consistent with the multiplicative inequality and the observed binary choice proportions. The multiplicative inequality on T reduces to $\hat{P}_T(c) \geq \hat{P}_{\{c,d\}}(c)\hat{P}_{\{c,e\}}(c)$ and $\hat{P}_T(d) \geq \hat{P}_{\{c,d\}}(d)\hat{P}_{\{d,e\}}(d)$ and $\hat{P}_T(e) \geq \hat{P}_{\{c,e\}}(e)\hat{P}_{\{d,e\}}(e)$. The vector $(\hat{P}_T(c), \hat{P}_T(d), \hat{P}_T(e))$ of ternary choice proportions lies outside the red triangle and in particular, $\hat{P}_T(d) = 0.35$ is less than $\hat{P}_{\{c,d\}}(d)\hat{P}_{\{d,e\}}(d) = 0.575 \times 0.65 = 0.37375$.

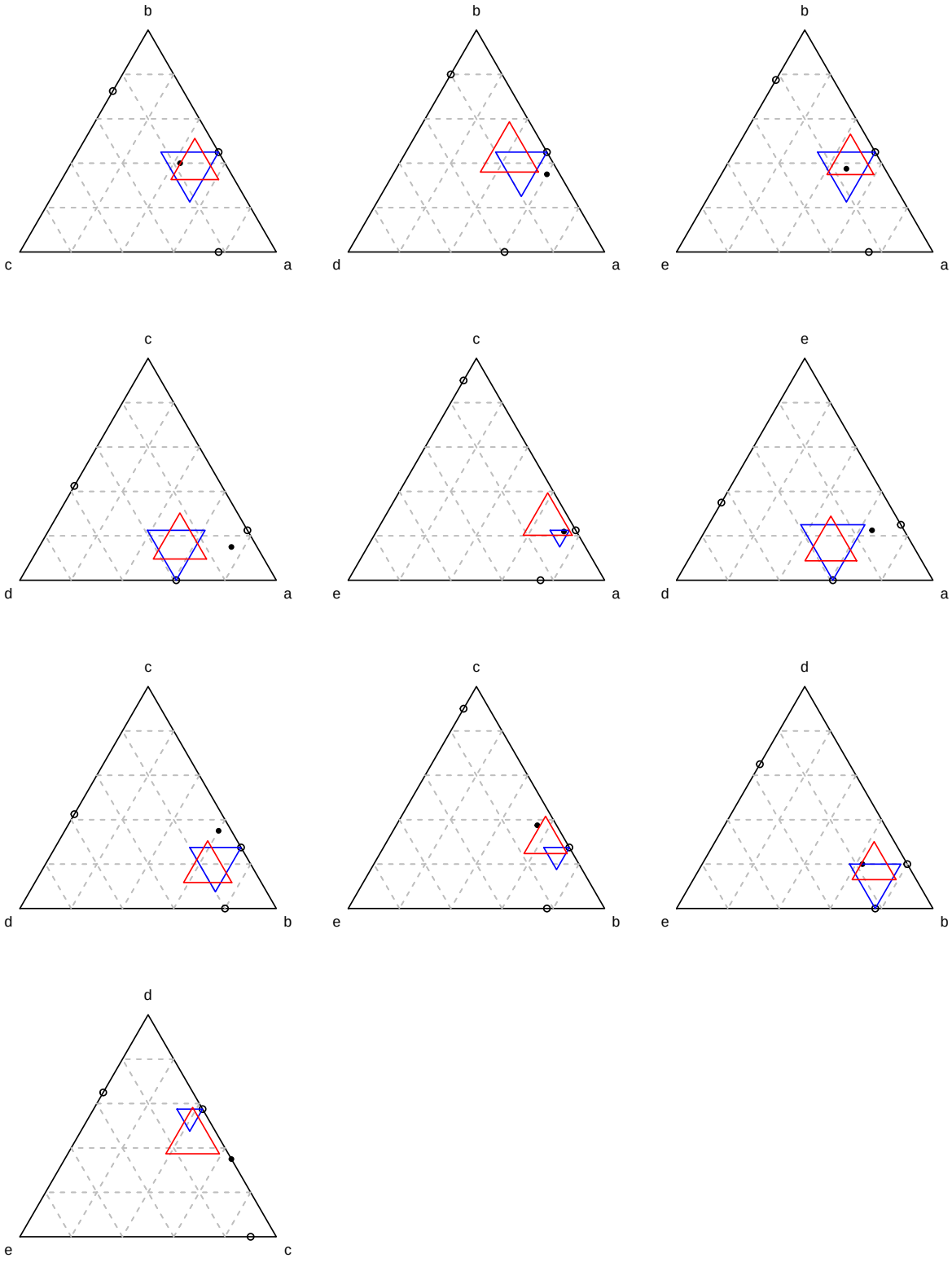
6 Transparency and openness

Supplementary materials, including data, code, experimental materials and other documents, are publicly available in the GitHub repository `https://github.com/mccauslw/Population`. The folder `data` includes the data from the experiment. The folder `R` contains R code used in the analysis. The folder `documents` includes the certificate of (ethics) approval, the consent form for participants, information on the recruitment and compensation procedures followed by SSI, the SSI Global Panel Book 2017, and participants' written feedback. Documents with figures analogous to Figure 1 and Figure 3 and tables analogous to Table 1, for each of the 32 domains, are in the folder `domains`. The authors declare no conflicts of interest. The study and analysis plan were not preregistered.

7 Conclusions

We have collected new experimental data featuring observations of population-level choice. Several features of the experimental design make the data suitable for testing abstract stochastic choice conditions at the population level. First, it includes a wide variety of choice domains. Some choice domains involve subjective aesthetic judgments, on such matters as food and drink, art, music, politics and the grammaticality of sentences. Other choice domains resemble choice domains that naturally occur in markets, such as those for mobile phone plans, hotel rooms, beer, cars and flights. Still others involve options having an objective rank order. Some choice domains have options that

Figure 3: Observed binary and ternary choice proportions for the "Latitude" domain. In each panel, hollow points represent binary choice proportions and the solid point represents ternary choice proportions. The downward- and upward-pointing triangles indicate the regions consistent with regularity and the multiplicative inequality, respectively. Analogous figures for all domains are in the supplementary materials.



are distinguished only by the levels of numerical attributes, others do not. Some domains are designed to be conducive to context effects, in order to stress-test conditions that are inconsistent with the effects. This variety helps us and others to discover properties of random choice structures that hold widely.

Second, for each choice domain, we observed repeated choices from all menus of a universe of five items, allowing us to test any condition by exposing every implication of it to empirical scrutiny. This is particularly important for testing conditions like regularity, random utility and the multiplicative inequality, where every menu-specific choice probability vector P_A is pivotal, in the sense defined in Section 2.2. Third, by arranging that every participant saw only one menu from each choice domain, we obtained data on population choices that are plausibly independent and, menu by menu, identically distributed.

The interpretation and generalizability of the results are necessarily tied to these design features. The data consist of population-level rather than individual-level choices, and implications for individual choice depend on the aggregation property discussed in Section 2.3. Participants were members of a Canadian online panel, so the results do not necessarily generalize to other populations, periods or elicitation methods. Although the 32 domains are deliberately diverse, they are not exhaustive.

The experimental dataset we have collected complements earlier work, including the abstract choice models introduced in McCausland & Marley (2013) and the inferential methods described in McCausland & Marley (2014). The data, supported by these models and methods, serve as a testing ground for the conditions we have considered in this paper and any others that one may wish to investigate. That said, the data are publicly available and stand alone, amenable to analysis using other models and methods.

Evidence in support of regularity and the random utility condition is fairly consistent across choice domains. While the evidence in favour of these conditions for any one choice domain is modest, the direction of the evidence is robust to the choice of prior and the consistency of the evidence across diverse domains constitutes important evidence in favour of the broad applicability of these conditions. While the literature on the attraction effect provides evidence against regularity and therefore random utility in special cases, these conditions hold up across the wide variety of choice domains we explore, including a few domains designed to be conducive to the attraction effect. This is reassuring given the prevalence of random utility models in the literature.

Evidence regarding the multiplicative inequality, in contrast, varies considerably across domains.

There is very strong evidence against it for three domains, so we can conclude that it cannot be considered as a universal condition for population-level stochastic choice. However, the multiplicative inequality does not have the aggregation property discussed in Section 2.3 so we cannot conclude that it is unsuitable for individual-level choice. Tversky (1972*b*) introduced the EBA class of models to accommodate the similarity effect, introduced in the same paper, but since the multiplicative inequality is a necessary condition for EBA models, there are population-level choice patterns that it cannot capture well. At the same time, the multiplicative inequality is favoured more than the random utility condition in many domains, which raises the question of what common features these domains might have. In future work we wish to investigate the multiplicative inequality in more detail, using a variety of data sources, complementing some theoretical work on that condition in McCausland & Marley (2024).

The dataset can be reused to evaluate other conditions on probabilistic choice behaviour, including behavioural axioms and conditions implied by different competing theories. A better understanding of choice behaviour requires evidence not only about specific anomalies, but also the broad applicability of such conditions. By providing a diverse population-level choice dataset, the present study aims to contribute to this endeavour.

Appendix A Prior distributions used in the prior sensitivity analysis

The prior distribution for P used for the three encompassing models is hierarchical: rather than choosing fixed values for α and λ, which is quite inflexible, we specify a prior for them in turn. Because results are somewhat sensitive to the choice of prior for λ (although less so to the choice of α) we did a prior sensitivity analysis.

Table 4 gives the exact specification of the prior distributions of α and λ for the encompassing models M_1, M_2 and M_3 used in the prior sensitivity analysis in Section 5.2. Each prior is defined in terms of the value of three tabulated hyper-parameters, a_1, a_2 and b. The table also provides the prior mean, variance and standard deviation of α and λ for each of the encompassing models.

In each case, α and λ are *a priori* independent. The prior distribution of the concentration parameter α is Gamma with shape parameter $a_1 + a_2$ and scale parameter b. The prior distribution of the dependence parameter λ is Beta with shape parameters a_1 and a_2.

Moving from M_1 to M_3 increases prior concentration near $\lambda = 1$, favouring progressively

	a_1	a_2	b	$E[\alpha]$	Var[α]	σ_α	$E[\lambda]$	Var[λ]	σ_λ
M_1	2.00	1.00	3.00	9.00	27.00	5.196	0.667	0.0556	0.236
M_2	2.50	0.50	3.00	9.00	27.00	5.196	0.833	0.0347	0.186
M_3	2.75	0.25	3.00	9.00	27.00	5.196	0.917	0.0191	0.138

Table 4: For the three encompassing models M_1, M_2 and M_3, the table shows the values of three hyper-parameters, a_1, a_2 and b, defining the prior distribution of α and λ; the prior mean, variance and standard deviation of α; and the same prior moments of λ.

stronger dependence across menus, while holding the prior distribution of α constant.

Appendix B Domains

We describe here the complete set of 32 choice domains featured in the experiment. We classify choice domains into four categories. In the first category, items have no numerical attributes. In the second category, items have a single attribute, whose level is not explicitly given. In the third category, items have two numerical attributes and the experimental design is intended to be conducive to one or more context effects. In the fourth category, items have many attributes, and they are chosen to resemble items in discrete choice experiments used in Marketing.

For many of the domains, we identify candidate dominance, similarity and betweenness relations. We designed these domains with these relations in mind, so that some configurations of stimuli would be conducive to attraction, similarity and compromise effects, respectively.

The section title ("Male stars" for example) for each domain gives the name of the domain for the purpose of reporting results. The specification of a domain is shown in a grey box and consists of a question and five choice items. In the actual experiment, for a given domain, all participants see the same question (such as "Which movie star would you choose to have lunch with?") but not the same set of options; different participants see different menus, consisting of from two to five of the five choice items, in random order, as described in Section 3. For the purpose of reporting results, the choice items in each domain are labelled a, b, c, d and e, in the order presented in the grey boxes. Because participants do not see these labels, we do not include them in the boxes.

We adopt the convention that whenever there is a natural order for the items in a list, we put them in decreasing order. Again, this is only for the purposes of reporting results; each participant sees the items in a menu in random order.

B.1 No numerical attributes

Many of the domains in this section were selected primarily to provide a diverse collection of straightforward and familiar choice environments without explicit numerical attributes.

B.1.1 Male stars

The source for this domain is the website `ranker.com`, accessed June 4, 2017. The list is "The best actors working today". The choices are the top five actors in that list, in order.

> Which movie star would you choose to have lunch with?
> - Tom Hanks
> - Kevin Spacey
> - Morgan Freeman
> - Leonardo DiCaprio
> - Christian Bale

B.1.2 Female stars

The source for this domain is the website `ranker.com`, accessed June 4, 2017. The list is "The best American actresses working today". These are the top five actors in that list, in order. Jodie Foster's name was misspelled in the experiment, as two participants noted in the comments.

> Which movie star would you choose to have lunch with?
> - Meryl Streep
> - Jody Foster
> - Kathy Bates
> - Amy Adams
> - Julianne Moore

B.1.3 Films

The source for this domain is the IMDb list "Most Popular Feature Films Released 1990 to 1999". The decade was chosen so that the films would not be easily recognizable by most respondents.

Judging from the following descriptions of films, which one of the films would you choose to see?

- Two imprisoned men bond over a number of years, finding solace and eventual redemption through acts of common decency.
- Mathilda, a 12-year-old girl, is reluctantly taken in by Léon, a professional assassin, after her family is murdered. Léon and Mathilda form an unusual relationship, as she becomes his protégé and learns the assassin's trade.
- The lives of two mob hit men, a boxer, a gangster's wife, and a pair of diner bandits intertwine in four tales of violence and redemption.
- A sexually frustrated suburban father has a mid-life crisis after becoming infatuated with his daughter's best friend.
- Identical twins, separated at birth and each raised by one of their biological parents, discover each other for the first time at summer camp and make a plan to bring their wayward parents back together.

B.1.4 Star pairs

Here, choice items are pairs of movie stars from a set of four stars: Tom Hanks, Scarlett Johansson, Brad Pitt and Angelina Jolie. We propose a similarity relation where two star pairs are similar if they have a star in common. The only star pairs that are not similar are c and d.

Knowing only who is starring, which one of these new films would you choose to see?

- Tom Hanks and Scarlett Johansson
- Scarlett Johansson and Brad Pitt
- Tom Hanks and Brad Pitt
- Scarlett Johansson and Angelina Jolie
- Tom Hanks and Angelina Jolie

B.1.5 Pizzas

The source for this domain is a Montreal pizza restaurant. All these pizzas are either 12 or 13 dollars.

Which one of the following pizzas would you choose?

- Mozzarella, tomato sauce, basil
- Pepperoni, mushrooms, green pepper, mozzarella, tomato sauce
- Red onion, tomato sauce, feta, mozzarella, olive oil, Greek spices, tomato sauce
- Bacon, white onion, mozzarella, parmesan, fresh cream, tomato sauce, ground pepper
- Mushrooms, green pepper, mozzarella, tomato sauce

B.1.6 Juices

Which one of the following fresh juices would you choose?

- Mango
- Orange
- Apple
- Grapefruit
- Pineapple

B.1.7 Colours

Which one of the following colours do you like best?

- Red
- Purple
- Pink
- Blue
- Green

B.1.8 Colour combos

The source for this domain is the website "The top tens", page "Two colors that look good side by side." The color combinations here are ranked 1, 4, 5, 13 and 14. We chose a set of combinations with many colors in common. We propose a similarity relation where two colour combinations are similar if they have a colour in common. All pairs of combinations are similar except a and e, and b and d. The name for this domain used in data files and documents, "Colour combos" is an abbreviation of the title seen by participants, "Colour combinations".

Which one of these colour combinations do you like best?

- Black and red
- Black and purple
- Black and blue
- Blue and red
- Blue and purple

B.1.9 Events

This domain involves comparisons of the probabilities of possible future events. Logically, the probability of event e must be at least as great as the probability of a, which must in turn be at

least as great as the probability of d; also, the probability of b must be at least as great as the probability of c. We propose a dominance relation based on these objective probability rankings. The dominance relation is somewhat unconventional, as most experimental designs in the literature on attraction effects feature numerical dominance relations.

Which one of the following events do you think is most likely to happen in the next twenty years?

- Scotland becomes an independent country.
- Either Catalonia or Quebec become independent countries.
- Catalonia becomes an independent country.
- Scotland and Quebec become independent countries.
- Either Scotland or Quebec become independent countries.

B.1.10 Radio formats

The domain is radio formats, and the choice items are the top 5 radio formats in Canada in 2015, according to

https://byrnesmedia.com/2015/10/05/the-6-best-performing-radio-formats-in-canada/

Descriptions of formats are from

http://www.newsgeneration.com/broadcast-resources/guide-to-radio-station-formats/

Suppose you were on a two hour road trip and you have a choice among radio stations with the following formats. Which one would you choose?

- News
- Hot Adult Contemporary, or Hot AC (A variety of classic and contemporary mainstream music geared towards adults.)
- Classic Hits (Rock and pop, roughly 1964-1989)
- Country Music
- Adult Contemporary, or AC (Adult-oriented pop/rock with no hard rock.)

B.1.11 Music

The choice items in this domain are the top selling musical artists of all time, according to Wikipedia. They should be familiar to a large majority of respondents. The name for this domain used in data files and documents, "Music" is an abbreviation of the title seen by participants, "Musical artists".

Which one of the following musical artists do you like the best?

- The Beatles
- Elvis Presley
- Michael Jackson
- Madonna
- Elton John

B.1.12 Aboriginal

The name for this domain used in data files and documents, "Aboriginal" is an abbreviation of the title seen by participants, "Aboriginal art".

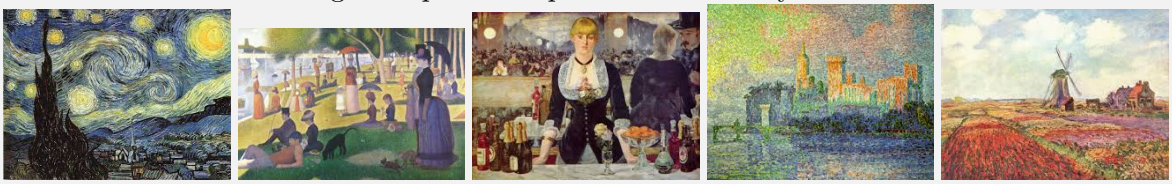
Which one of the following examples of Australian aboriginal art do you like the best?



B.1.13 Impressionist

The name for this domain used in data files and documents, "Impressionist" is an abbreviation of the title seen by participants, "Impressionist art".

Which one of the following examples of Impressionist art do you like the best?



B.1.14 Sentences

This domain consists of sentences that have been used in experiments of linguistic judgment. They are, respectively sentences 7j, 7h, 7p, 7a and 7e in Bard et al. (1996). Figure 1 of that paper shows acceptability scores for these and other sentences given by two individual linguists, an acceptability score aggregating the scores of four linguists and an acceptability score aggregating the scores of four "naive respondents", all undergraduate anatomy students. There is broad, but not perfect,

agreement in terms of order, and in the following list, they are in decreasing order of acceptability according to the measure aggregating the judgments of four linguists.

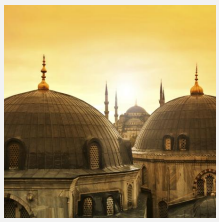
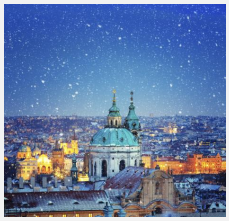
Which one of the following sentences do you find the most grammatically acceptable?

- Who did Bill buy the car to please?
- This is a book which reading would be fun.
- Where did Bill buy the car to drive?
- Which man do you wonder when to meet?
- With which pen do you wonder what to write?

B.1.15 Travel

The source is Tripadvisor. These are the top five travel destinations, according to the results of an on-line contest where visitors to a Tripadvisor site could make pairwise choices between travel destinations.

Which one of the following travel destinations would you most like to visit?

1. Marrakech, Morocco 
2. Istanbul, Turkey 
3. Hanoi, Vietnam 
4. Siem Reap, Cambodia 
5. Prague, Czech Republic 

B.1.16 Marijuana

This question elicits policy preferences.

Which one of the following marijuana policies would you choose?

- Possession by, and sales to adults are both legal; sales to minors are illegal.
- Possession by, and sales to adults are both illegal but neither is a criminal offense; sales to minors are a criminal offense.
- Possession is illegal but not criminal; all sales are a criminal offense.
- Possession and sales are criminal offenses, with a small number of medical exceptions.
- Possession and sales are criminal offenses, without exception.

B.2 Single attributes, not directly observed

The five domains in this section contain choice items with an objective rank order.

B.2.1 Latitude

The five cities of this domain have a latitude close to 50 degrees north. In the following list, they are ordered from furthest north to furthest south. According to Wikipedia, their latitudes are, respectively, $52°14'N$, $51°30'N$, $49°15'N$, $48°51'N$ and $47°36'N$. We propose a dominance relation in which Vancouver dominates Seattle and London dominates Paris, based on the conjecture that participants will more easily discriminate between cities of similar longitude.

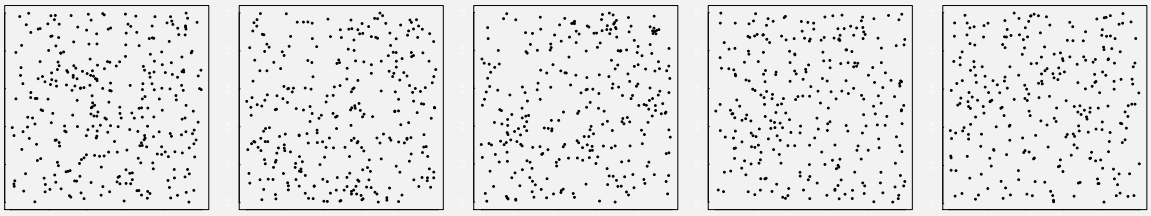
Which one of the following cities do you think is furthest north?

- Warsaw, Poland
- London, United Kingdom
- Vancouver, Canada
- Paris, France
- Seattle, United States

B.2.2 Dots

This domain is a perception example. The true numbers of points are, respectively, 320, 310, 300, 290 and 280. We propose a similarity relation where two options are similar if the difference in the number of points is only 10.

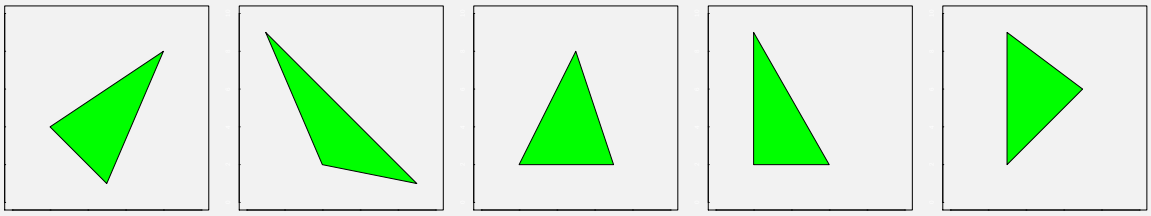
Which one of the following boxes do you think has the greatest number of points?



B.2.3 Triangles

This domain is another perception example. The true areas are, respectively, 16, 15, 15, 14 and 14 units.

Which one of the following triangles do you think has the greatest area?



B.2.4 Population

These countries were ranked, respectively, 4th through 8th in terms of population in 2016, when their populations, in millions, were 258, 206, 202, 186 and 156.

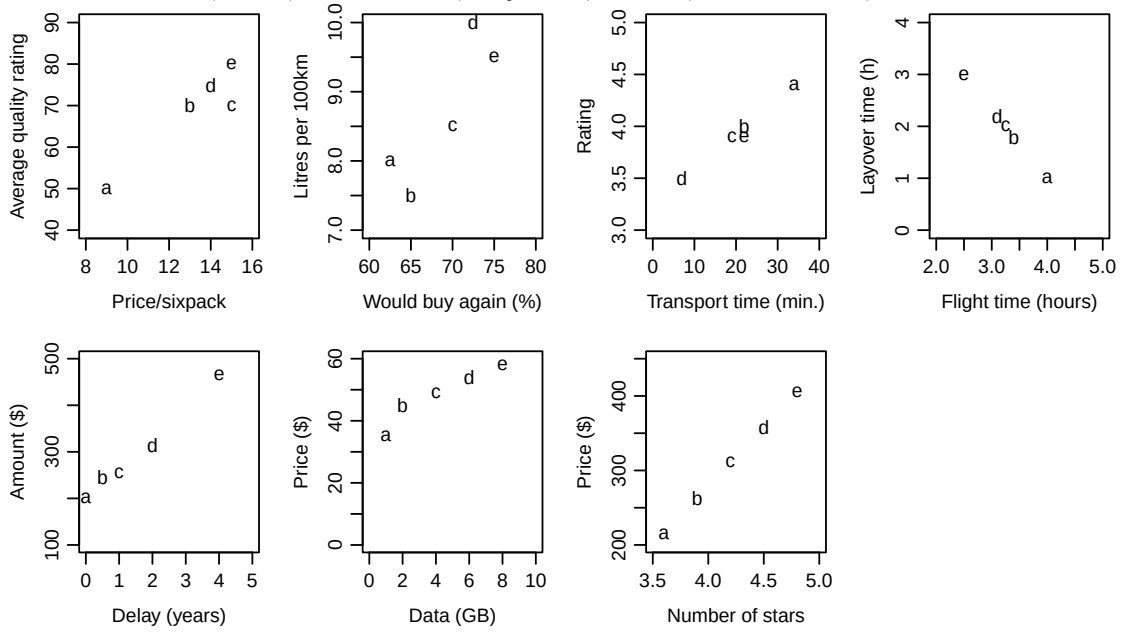
Which one of the following countries do you think has the largest population?

- Indonesia
- Brazil
- Pakistan
- Nigeria
- Bangladesh

B.2.5 Surface area

These countries are ranked, respectively, 2nd through 6th in terms of surface area, including inland bodies of water. In millions of square kilometres, those surface areas are 9.984, 9.573, 9.525, 8.516 and 7.692.

Figure 4: Designs for context effects. Each panel plots the five items of a domain in attribute space. The domains are Beer, Cars, Restaurants, Layovers, Future, Phone Plans, and Hotels.



Which one of the following countries do you think has the greatest surface area, including inland bodies of water?

- Canada
- United States of America
- China
- Brazil
- Australia

B.3 Items with two attributes

Here we describe domains whose choice items have exactly two real-valued attributes. In all cases, the levels of these attributes are presented directly to the participant. Figure 4 plots the five items of each of these domains in attribute space, revealing candidate relations of dominance, similarity and betweenness.

Table 5 shows, for the first four of these domains, candidate dominance, similarity and betweenness relations. Following usual practice, we exclude similarity relations for two items for which there is also a dominance relation between the two; and betweenness relations among three items when two of them are similar and the third is not similar to either of the first two.

Domain	Dominance	Similarity w/o dominance	Betweenness w/o similarity
Beer	$b > c$, $d > c$, $e > c$	$b \sim d$, $d \sim e$	$a\|b\|e$
Cars	$b > a$, $e > d$		$a\|c\|e$, $b\|c\|e$, $a\|c\|d$, $b\|c\|d$
Restaurants	$b > e$, $c > e$	$b \sim c$	$a\|b\|d$, $a\|c\|d$, $a\|e\|d$
Layovers		$b \sim c$, $c \sim d$, $b \sim d$	$a\|b\|e$, $a\|c\|e$, $a\|d\|e$

Table 5: Candidate relations of dominance, similarity and betweenness in two-attribute domains. For items x, y and z, $x > y$ means that x dominates y, $x \sim y$ means that x and y are similar, and $x|y|z$ means that y is between x and z, in the sense defined in Section 2.4.

B.3.1 Beer

This domain is from an experiment reported in Huber et al. (1982) used to illustrate an attraction effect. The prices are multiplied by 5 and we added choice items d and e to allow for more configurations conducive to attraction and similarity effects. The most commonly used domains to illustrate the attraction effect involve beer, cars, apartments, computers, restaurants and televisions.

Below you will find three brands of beer. You know only the price per sixpack and the average quality ratings made by respondents in a blind taste test. Given that you had to choose one brand to buy on this information alone, which one would you choose?

Price/sixpack	Average quality rating (100 = Best; 0 = Worst)
\$9.00	50
\$13.00	70
\$15.00	70
\$14.00	75
\$15.00	80

B.3.2 Cars

This domain is based on an experiment from Wedell & Pettibone (1996), numbered 18 in the appendix to that paper. Items below have the same attributes as in that experiment and a similar range of attribute levels. We adapted the items to include more betweenness configurations.

Which one of the following similarly priced cars would you choose to drive, based on the percentage of owners who indicated that they "would buy again," and the fuel economy rating?

Percentage responding "would buy again"	Fuel economy rating in $L/100\,\text{km}$
62.5	8.0
65.0	7.5
70.0	8.5
72.5	10.0
75.0	9.5

B.3.3 Restaurants

This domain is based on another experiment in Wedell & Pettibone (1996), numbered 19 in the appendix to that paper.

Which one of the following restaurants would you choose for your next restaurant meal, based on transportation time (in minutes) and average customer ratings (from 1 to 5).

Transportation time	Rating
34	4.4
22	4.0
19	3.9
7	3.5
22	3.9

B.3.4 Layovers

The name for this domain used in data files and documents, "Layovers", is an abbreviation of the title seen by participants, "Flight layovers".

Which one of the following flight itineraries would you choose? All involve two flights, with one layover between them.

Total inflight time	Layover time	Total itinerary time
4:00	1:00	5:00
3:24	1:48	5:12
3:15	2:00	5:15
3:06	2:12	5:18
2:30	3:00	5:30

B.3.5 Future

This domain is loosely based on an experiment by Benzion et al. (1989), in which respondents are asked to assign present values equivalent to the receipt of $200 at time horizons of 0.5, 1, 2 and 4 years. Based on implied discount factors at various terms, we constructed five choice items designed to have approximately the same present value equivalent. The name for this domain used in data files and documents, "Future", is an abbreviation of the title seen by participants, "Future payments".

Which one of the following would you choose?

- $200 credited to your bank account immediately.
- $245 credited to your bank account in six months.
- $255 credited to your bank account in one year.
- $315 credited to your bank account in two years.
- $465 credited to your bank account in four years.

B.3.6 Phone plans

The source for this domain is the website of Fido Mobile, a Canadian cell phone carrier, with rates quoted on March 1, 2017 in Canadian dollars.

Of the following cell phone plans, which one would you choose? In all cases, unlimited calling, text picture and video messages to Canadian and international mobile numbers are included. Excess data usage is billed at $10 per 500 MB.

- 1 GB data per month, $35 per month.
- 2 GB data per month, $45 per month.
- 4 GB data per month, $49 per month.
- 6 GB data per month, $54 per month.
- 8 GB data per month, $58 per month.

B.3.7 Hotel rooms

Using Expedia results, we did a linear regression of price per night on a constant and the Expedia rating, in numbers of stars, for a sample of available hotels. The five levels of numbers of stars correspond roughly to the mean, plus and minus one sample standard deviation and plus and minus two standard deviations. Prices are approximately equal to fitted values in the linear regression.

Suppose you are staying over two nights in New York city. Which one of the following hotels would you choose, based on customer ratings and price per night?

- 3.6/5 stars, $215 per night
- 3.9/5 stars, $263 per night
- 4.2/5 stars, $311 per night
- 4.5/5 stars, $358 per night
- 4.8/5 stars, $406 per night

B.4 Items with more than two attributes

B.4.1 Itineraries

This domain illustrates three-way tradeoffs. The points form a constellation in a plane that resembles the pattern of points on the "five" side of a die. The name for this domain used in data files and documents, "Itineraries", is an abbreviation of the title seen by participants, "Two-flight itineraries".

Which one of the following flight itineraries would you choose? All involve two flights and have a total duration of six hours.

1st flight	Layover	2nd flight
1:30	1:15	3:15
3:15	1:15	1:30
2:15	1:30	2:15
1:30	1:45	2:45
2:45	1:45	1:30

B.4.2 Televisions

The source for this domain is the website of Best Buy Canada, with prices in Canadian dollars.

Which one of the following televisions would you choose to buy if you were in the market for a television? All are LED televisions. Resolution refers to number of horizontal lines. Smart indicates internet connectivity.

Brand	Resolution	Smart	Price ($)	Screen Size (inches)
Sharp	1080	Yes	309	32
Insignia	720	No	209	32
Sony	720	Yes	439	32
Samsung	1080	Yes	459	40
Toshiba	1080	No	409	43

B.4.3 Coffee

The source for this domain is the website `buycoffeecanada.com`, with prices in Canadian dollars.

You need to buy 16oz of ground coffee for a brunch with friends. Which one of the following ground coffees would you choose?

Price	Fair Trade	Name: Description
18.71	Yes	Ethiopian Yirgacheffe: vibrant and intensely aromatic, fruity
9.99	No	Colombian Supremo: mellow cup, complex aromas and rich flavours
13.72	Yes	Colombian Organic: medium body, fragrant aroma and mild acidity
12.35	No	Tanzania Peaberry: full bodied coffee, chocolatey aroma, wine-like finish
13.46	No	Sumatra Mandheling: exotic, earthy, bright with low acidity

B.4.4 Charity

The charities in this domain are real charities. The first two have broadly supported health-related missions. However, the Canadian Cancer Society attracts much more financial support than Arthritis Research Canada and we might expect that most people would prefer a marginal dollar going to the former. The two other charities have goals that are nearly opposite and elicit strong emotions (of different kinds) from some.

Suppose someone was donating a total of 100 dollars to a combination of charities, on your behalf. Which one of the following divisions of the 100 dollars would you choose?

Arthritis Research Canada	Canadian Cancer Society	Canadian Coalition for Firearm Rights	Coalition for Gun Control
90	10	0	0
35	60	5	0
35	60	0	5
10	80	10	0
10	80	0	10

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